

Problem 1: The move: take one counter from every pile and put all the counters you took together as a new pile. A pile with no counters left is gone. Write an arrangement biggest pile first, like 4-2. Run these four starts until the arrangement stops changing, one start on each strip, and write the number of moves each took: 6; 3-3; 4-1-1; 2-2-1-1. Then, for every move of the 2-2-1-1 run, write in the table the arrangement before the move, its number of piles, and the size of the new pile the move made.

6	→		→		→		→	
→		→		→			moves: _____	
3-3	→		→		→		→	
→		→		→			moves: _____	
4-1-1	→		→		→		→	
→		→		→			moves: _____	
2-2-1-1	→		→		→		→	
→		→		→			moves: _____	

before the move	number of piles	size of the new pile

Problem 2: List every way to split six counters into piles, biggest pile first. There are eleven. Sort them by biggest pile to be sure none is missing. Then make the map on the next page: 3-2-1 is already in its box at the bottom. For each other arrangement, draw a box above the arrangement it becomes after one move, write it in the box, and draw an arrow from the box to that arrangement. When the map is finished, check every arrangement: does its path of arrows lead to 3-2-1?

- | | |
|----------|-----------|
| 1. _____ | 7. _____ |
| 2. _____ | 8. _____ |
| 3. _____ | 9. _____ |
| 4. _____ | 10. _____ |
| 5. _____ | 11. _____ |
| 6. _____ | |

3-2-1



Problem 3: From your map, write in the table how many moves each of the eleven arrangements needs to reach 3-2-1. Which needs the most? How many arrangements need exactly three moves? Which arrangements have no arrow pointing at them? Which arrangements have two arrows pointing at them?

arrangement	moves

arrangement	moves

most moves: _____

exactly three moves: _____ arrangements

no arrow in: _____

two arrows in: _____

Problem 4: Build 3-2-1 and make the move. Write what happened to the pile of 3, to the pile of 2, to the pile of 1, and how big the new pile is. Now suppose an arrangement does not change when you make the move. After the move, is any old pile as big as it was before? So which pile after the move must be the new pile: the biggest, the smallest, or one in between? Write which, and why. Then write what the biggest pile, the next, and the next must be, down to the smallest pile. Finally write every number of counters from 1 to 30 that has an arrangement which does not change.

the pile of 3 became: _____

the pile of 2 became: _____

the pile of 1: _____

the new pile has: _____

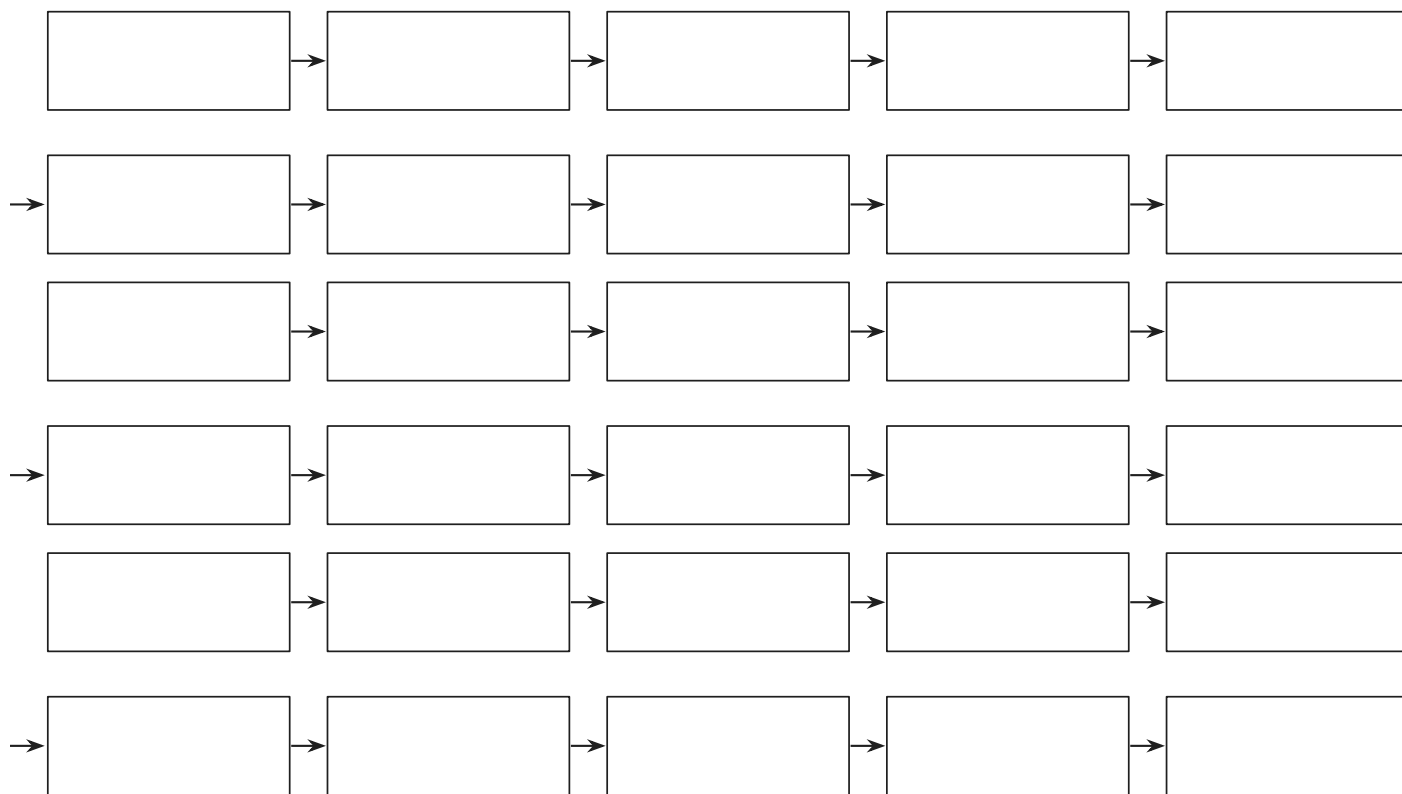
the new pile must be

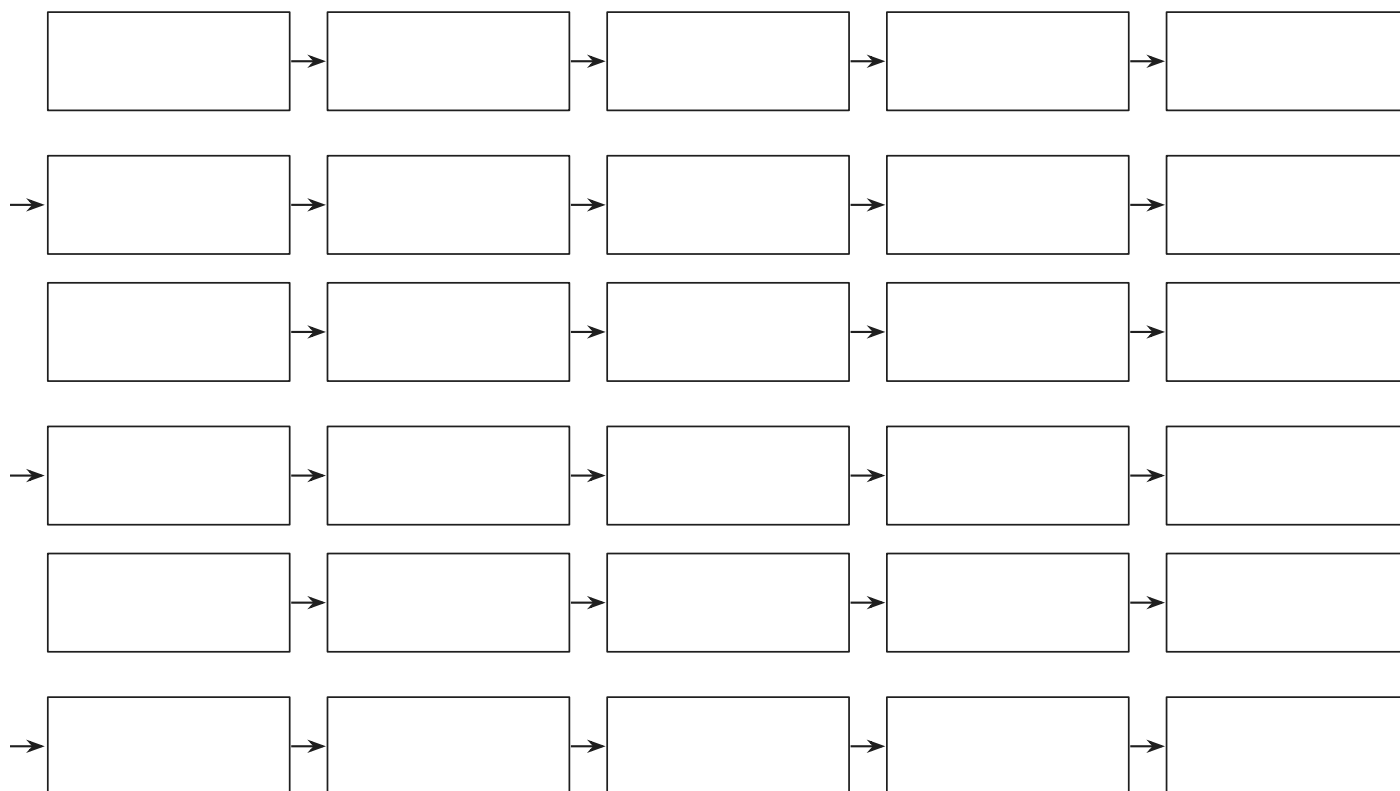
the piles must be

numbers from 1 to 30: _____

Problem 5: List all fifteen ways to split seven counters into piles, biggest pile first. Split them between you. Run each one until an arrangement appears a second time. Then draw the map of seven on the blank page: every arrangement in a box, with an arrow from it to what it becomes. Circle the loop. How many arrangements are in the loop? How many moves does the farthest start need to enter it? Which arrangements have no arrow pointing at them?

- | | |
|----------|-----------|
| 1. _____ | 9. _____ |
| 2. _____ | 10. _____ |
| 3. _____ | 11. _____ |
| 4. _____ | 12. _____ |
| 5. _____ | 13. _____ |
| 6. _____ | 14. _____ |
| 7. _____ | 15. _____ |
| 8. _____ | |





arrangements in the loop: _____

farthest start: _____ needs _____ moves

no arrow in: _____

Problem 6: Run 8, 5-3, 4-4, 2-2-2-2 and 6-1-1 until each one repeats an arrangement, and circle each loop. How many different loops did you find? Write the arrangements of each loop as a ring in a box of its own on the next page, in the order they come. How long is each loop? Which of the five starts land in which loop?

	8	→		→		→		→	
→		→		→		→		→	
	5-3	→		→		→		→	
→		→		→		→		→	
	4-4	→		→		→		→	
→		→		→		→		→	
	2-2-2-2	→		→		→		→	
→		→		→		→		→	
	6-1-1	→		→		→		→	
→		→		→		→		→	

a loop



a loop



a loop

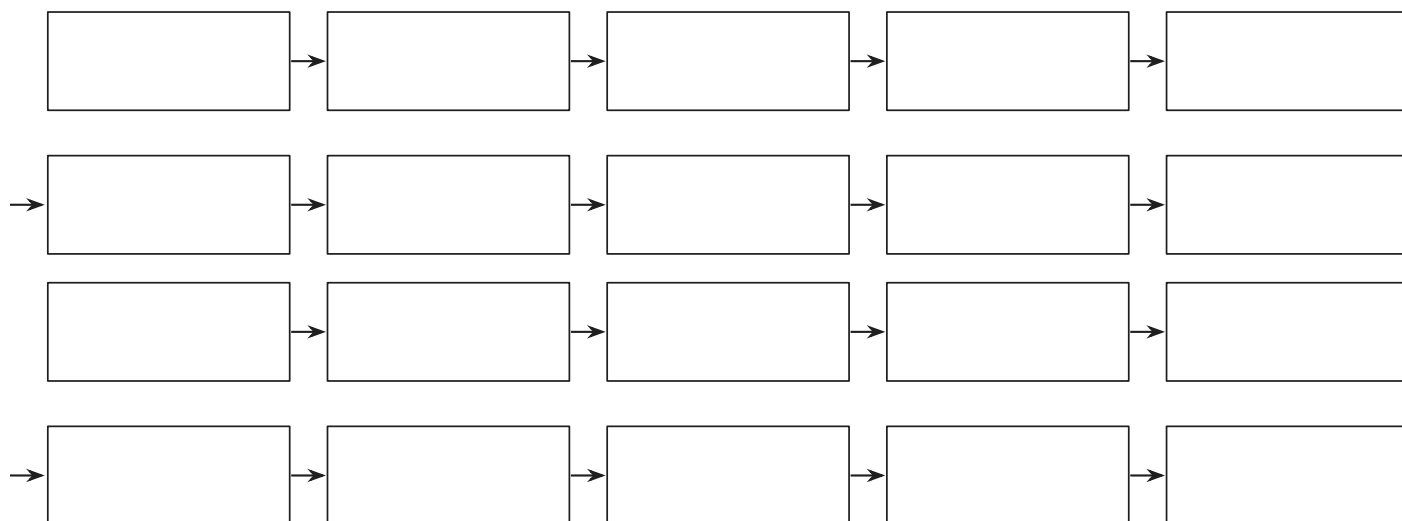


different loops: _____

lengths: _____

Problem 7: List all twenty-two ways to split eight counters into piles, biggest pile first. Next to each one, write the loop it reaches: most of them are already on your strips from Problem 6; run the rest. Count how many starts reach each loop.

	loop		loop
1. _____	_____	12. _____	_____
2. _____	_____	13. _____	_____
3. _____	_____	14. _____	_____
4. _____	_____	15. _____	_____
5. _____	_____	16. _____	_____
6. _____	_____	17. _____	_____
7. _____	_____	18. _____	_____
8. _____	_____	19. _____	_____
9. _____	_____	20. _____	_____
10. _____	_____	21. _____	_____
11. _____	_____	22. _____	_____



starts reaching each loop: _____

Problem 8: An arrangement that becomes X after one move is called a parent of X . Find every parent of 5-3-2 (ten counters). Then every parent of 4-4 (eight counters). Then every parent of 2-2-2-2. Write them. Then describe a way to find every parent of any arrangement without guessing.

parents of 5-3-2: _____

parents of 4-4: _____

parents of 2-2-2-2: _____

Problem 9: From the map of six, list the arrangements with no arrow pointing at them, and from the map of seven do the same. For each of them write the biggest pile and the number of piles in the table. Find a rule that tells, from these two numbers, whether an arrangement has a parent. Write the rule. Use it to list every arrangement of eight counters with no parent. Then check two of them by trying to build a parent.

arrangement	biggest pile	number of piles

rule

eight counters, no parent

- 1. _____
 - 2. _____
 - 3. _____
 - 4. _____
 - 5. _____
 - 6. _____
 - 7. _____
 - 8. _____
-
-

Problem 10: Run 10, 5-5, 7-3 and 4-4-1-1 until they stop changing and write the number of moves for each. Then search for a start with ten counters that takes twelve moves. Write its whole run.

10	→		→		→		→	
----	---	--	---	--	---	--	---	--

→		→		→		→		→	
---	--	---	--	---	--	---	--	---	--

→		→		moves: _____
---	--	---	--	--------------

5-5	→		→		→		→	
-----	---	--	---	--	---	--	---	--

→		→		→		→		→	
---	--	---	--	---	--	---	--	---	--

→		→		moves: _____
---	--	---	--	--------------

7-3	→		→		→		→	
-----	---	--	---	--	---	--	---	--

→		→		→		→		→	
---	--	---	--	---	--	---	--	---	--

→		→		moves: _____
---	--	---	--	--------------

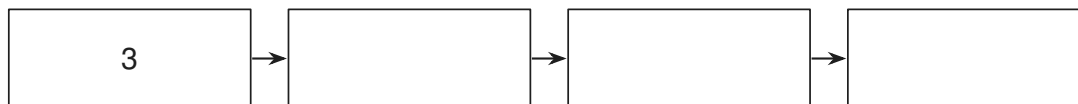
4-4-1-1	→		→		→		→	
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→		→		→		→		→	
---	--	---	--	---	--	---	--	---	--

→		→		moves: _____
---	--	---	--	--------------

→				
→				
→				
→				
→				
→				

Problem 11: Run one pile of three counters until it stops changing. Look back at how many moves one pile of six took (Problem 1) and one pile of ten took (Problem 10). Write your prediction for one pile of fifteen. Then run it and write every arrangement. In each arrangement of your run, underline the biggest pile and look at the rest. What do the small piles look like after 1, 3, 6 and 10 moves? Then predict how many moves 1-1-1-1-1-1-1-1-1-1-1-1-1-1-1 takes and write why.



moves: _____

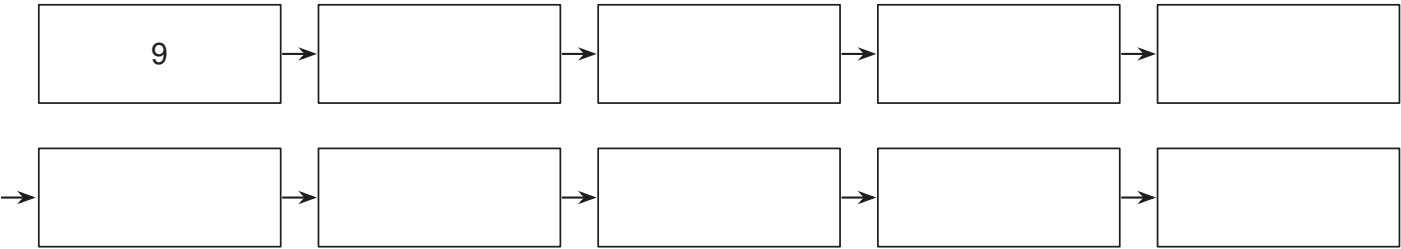
prediction for fifteen: _____



moves: _____

fifteen ones: _____ moves

Problem 12: Run nine counters in one pile until an arrangement appears a second time, and circle the loop. Copy every loop you have found, for 7, 8 and 9 counters, into the table, one arrangement per row. Try to write each loop arrangement as the staircase 3-2-1 with extra counters added, at most one extra on each step and at most one extra as a new pile of 1, like $4-2-1 = (3+1)-2-1$ and $3-2-1-1 = 3-2-1-(0+1)$. Does it work for every loop arrangement? Follow the extras around each loop: where does each extra go after one move? Use that to explain why the loop for 7 has four arrangements, why 8 has two loops of different lengths, and why 9 has one loop of four.



counters	loop arrangement	as 3-2-1 plus extras

Problem 13: Lay out six playing cards face up as the staircase 3-2-1: a pile of three on the left with the ace on top, then the 2, then the 3 at the bottom; a pile of two in the middle with the 4 on top of the 5; the 6 alone on the right. The move with cards: take the top card of every pile. Stack the cards you took, the card from the right pile at the bottom, then the card from the middle pile, then the card from the left pile on top. Put this new pile on the left and slide the other piles one place to the right. After each move, write in the table which cards are in each pile, top to bottom. Make six moves. Which card never moves? Which cards are back in their starting place after two moves? After three? After how many moves is every card back where it started? Then lay out ten cards as 4-3-2-1 (ace, 2, 3, 4 on the left, top to bottom; then 5, 6, 7; then 8, 9; then 10) and find after how many moves every card is back.

	left	middle	right
after 0 moves	A 2 3	4 5	6
after 1 move			
after 2 moves			
after 3 moves			
after 4 moves			
after 5 moves			
after 6 moves			

never moves: _____

back after 2 moves: _____

back after 3 moves: _____

all back after: _____ moves

	left	middle	right	far right
after 0 moves	A 2 3 4	5 6 7	8 9	10
after 1 move				
after 2 moves				
after 3 moves				
after 4 moves				
after 5 moves				
after 6 moves				
after 7 moves				
after 8 moves				
after 9 moves				
after 10 moves				
after 11 moves				
after 12 moves				

all ten back after: _____ moves